AAKASH S

Full Stack Python Developer | Machine Learning Enthusiast

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Full Stack Python Developer and **Machine Learning** enthusiast pursuing MCA at the University of Madras, experienced in building end-to-end ML applications with up to **99%**+ model accuracy using **Python**, **Flask**, **FastAPI**, **SQL**, **Scikit-learn**, and **Pandas**. Proficient in **object-oriented programming**, **REST API development**, and Docker-based deployment, with growing exposure to **React.js** and full stack workflows. Seeking to contribute as a Software Engineer in full stack or applied ML roles.

TECHNICAL SKILLS

- **Programming languages :** Python, C++
- **Frontend** : HTML, CSS, JavaScript, React.js
- **Backend:** FastAPI, Flask, REST APIs
- **Machine Learning:** Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
- **Database:** SQL, MySQL
- **Tools & Technologies** : Git, GitHub, Docker

PROJECTS

Machine Learning-Based Transaction Risk Analysis

- Built a full-stack ML web application that classifies financial transactions as **Low, Medium, or High risk** using 15+ features including transaction amount, device type, IP risk score, and login attempts.
- Trained a Random Forest Classifier on ~1,000 transaction records, achieving **99.1% test accuracy**.
- Developed an end-to-end pipeline with **Flask, MySQL, Joblib**, including authentication, dashboard, transaction form, and real-time prediction.

AI Virtual Mouse using Hand Gestures

- Developed a real-time virtual mouse using **Python** and **OpenCV**, achieving ~50 FPS hand tracking for responsive cursor control.
- Implemented **gesture recognition** for 5 actions — left click, right click, scroll, zoom in, and zoom out — using hand landmark detection.
- Enabled **touchless computer interaction** by replacing physical mouse input with computer vision-based control.

Hybrid Neuro-Symbolic Machine Learning for Logical Reasoning Tasks (Sudoku solver)

- Developed a hybrid Sudoku solver combining symbolic backtracking with a **neural-guided prioritization heuristic**, reducing backtracking operations by **99.7% (1,718 → 6 average)** on 9×9 grids with 65% cells removed.
- Implemented **constraint validation** across rows, columns, and 3×3 boxes, achieving zero invalid **placements** across test runs.
- Benchmarked the hybrid approach against **pure backtracking** over 10 trials, demonstrating more efficient search despite comparable wall-clock time.

INTERNSHIPS

Vcodez

Machine Learning Intern Oct 2025 – Jan 2026

- Completed a hands-on internship in **Machine Learning and Data Science using Python**, covering the ML workflow from data preprocessing to deployment.
- Worked on **data preprocessing, visualization, and model development** using **Scikit-learn** on real-world datasets.
- Developed a **Streamlit**-based weather dashboard integrating multiple **OpenWeatherMap API** endpoints (current, forecast, air quality) with a rule-based advisory engine and dynamic weather-responsive UI.

ICT Academy & OpenWeaver Aug 2023 - Sep 2023

Data Science Intern

- Completed a hands-on Data Science and Machine Learning bootcamp covering fundamental concepts and applied ML workflows.
- Built a low-code e-commerce web application using drag-and-drop UI components and automated backend workflow logic.
- Integrated Gmail-based order notifications and configured order-management workflows, gaining exposure to UI design and deployment.

CERTIFICATIONS & WORKSHOPS

- **Cyber Security Workshop – Stella Maris College (2024)**
- **Mobile Application Development Workshop –** Prathyusha Engineering College (2024)
- **Infosys Springboard -** (Artificial Intelligence, Computer Vision 101, OpenAI GPT Models, Deep Learning, NLP) (2023)
- **Naan Mudhalvan –** Microsoft 360° Course (2023)
- **International Workshop on Data Science using Python –** SACAS & Brainovision Solutions (2023)

EDUCATION

Master of Computer Applications (MCA)- Pursuing (Distance Education) Chennai, Tamilnadu, India
University of Madras | Percentage: 67% July 2024 - Present

Bachelor's of Computer Applications (BCA)
S.A. College of Arts and Science | Percentage: 68% Aug 2021 - May 2024